

FITTRACK: GYM MEMBERSHIP & ATTENDANCE ANALYZER

A MINI PROJECT REPORT

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CS2307 - Internet Programming Laboratory

Submitted by

VISHAL K, III ADS (24UAD010)
BALA YOGHI V, III ADS (24UAD045)
PAVITHRAN, III ADS (24UAD056)

of

KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institution - Affiliated to Anna University, Chennai)
K.VELLAKULAM, VIRUDHUNAGAR - 625 701

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ABSTRACT

The project entitled “FITTRACK: GYM MEMBERSHIP & ATTENDANCE ANALYZER” is developed using HTML, CSS and JavaScript as the front end, PHP as the server side scripting language and MySQL as the back end. A gymnasium has to maintain the personal details of its members, the membership plan purchased by each member, the validity of that membership and the attendance of every member on every working day. When these records are kept in manual registers, the same member details are written again for every new membership, and finding how regularly a particular member attends the gym becomes a slow and error prone activity. The proposed system stores all this information in a single relational database and allows the gym administrator to manage it through an ordinary web browser.

The project consists of five modules: Member Registration, Membership Management, Attendance Management, Member Management and Report and Analysis. Member Registration module inserts the personal details of a new member such as name, age, gender, phone number, email, address and joining date into the members table, and a unique member identification number is generated for every member. Membership Management module assigns a membership plan to a registered member, records the start date and the end date of that membership and maintains the membership status as Active, Expired or Expiring Soon.

Attendance Management module records the daily attendance of a member by storing the attendance date, the check-in time and the check-out time, and the workout duration of that visit is obtained from the check-in and check-out time. Member Management module allows the administrator to search for an existing member, to modify the stored details when they change and to remove a member record when it is no longer required. Report and Analysis module reads the stored data and presents the daily attendance, the attendance of an individual member, the attendance percentage, the average workout duration and the list of active, expired and expiring memberships.

All the five modules are implemented using the four basic database operations, namely INSERT, SELECT, UPDATE and DELETE, which are executed from PHP through SQL queries submitted to the MySQL server. The attendance percentage of a member is obtained as the ratio of the number of days present to the total number of working days, expressed as a percentage, and the average workout duration is obtained as the ratio of the total workout duration to the total number of attendance days. The proposed system reduces manual record keeping, avoids the duplication of member information and gives the gym administrator a quick and reliable view of membership and attendance.

CHAPTER I

INTRODUCTION

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INTRODUCTION

A gymnasium is a service organisation in which the same set of members visits the premises repeatedly over a long period of time. The value that a member receives depends not only on the membership that has been purchased but also on how regularly the member actually attends. The management of a gym therefore has to keep two different kinds of records at the same time, namely the commercial record of who is a valid member and until when, and the operational record of who attended on which day. FitTrack has been developed to keep both of these records in one place and to derive useful information from them.

1.1 OVERVIEW

FitTrack: Gym Membership & Attendance Analyzer is a web based application that manages the members of a gymnasium, the membership plans offered by the gymnasium, the membership purchased by each member and the daily attendance of the members. The application is developed as a set of PHP pages which are accessed through a web browser. The pages are designed using HTML and CSS, simple validations on the entered data are performed using JavaScript, and the information entered by the administrator is stored in a MySQL database.

The administrator logs into the application and is presented with the operations of the system. A new member is registered by entering the personal details, after which a membership plan is assigned to that member and the start date and end date of the membership are recorded. On every working day the administrator marks the attendance of the members who visit the gym by recording the check-in time and the check-out time, from which the duration of the workout is obtained. The details of the members already present in the database can be searched, updated when they change, and deleted when the record is no longer required. Finally, the information that has been collected over a period of time is read back from the database and presented as attendance and membership reports.

1.2 MOTIVATION

Most small and medium gymnasiums still maintain their records in paper registers. One register holds the member details, another holds the fee and membership details, and a third register is kept at the entrance for attendance. Because the same member appears in all three registers, the same information is written repeatedly, and a change such as a new phone number has to be corrected in several places, which is rarely done. Searching for the record of a particular member

requires the pages of a register to be turned one by one, and the register of an earlier year is often not available at all.

The difficulty becomes greater when information has to be derived rather than merely retrieved. A question as simple as how many days a particular member attended during the last month, or which memberships are going to expire in the coming week, cannot be answered from a paper register without counting the entries by hand. As a result such questions are usually not asked at all, and the management loses the opportunity to remind a member whose membership is about to expire or to speak to a member who has stopped attending. These practical difficulties motivated the development of a computerised system in which the records are entered once, stored in a structured form and analysed automatically.

1.3 IMPORTANCE OF THE WORK

The importance of this work lies in bringing the membership record and the attendance record of a gymnasium into a single database so that each of them can be used to explain the other. Because the member details are stored only once and are referred to from the membership record and from the attendance record through the member identification number, the duplication that is unavoidable in a manual system is eliminated and the stored information remains consistent.

The work is also important because it converts the raw attendance entries into information that can actually be used. The number of days a member was present, the attendance percentage of that member and the average duration of the workouts are calculated by the system from the entries that have already been made, without any additional effort on the part of the administrator. In the same way, the membership status of every member is derived from the start date and the end date, so that the memberships that are active, the memberships that have expired and the memberships that are about to expire can be listed at any time. The system therefore replaces a collection of registers with an organised store of data and a set of reports drawn from it.

1.4 OBJECTIVES

The main objectives of the project are as follows:

- To store the personal details of the gym members in a centralised database instead of in manual registers.
- To assign a membership plan to a registered member and to maintain the start date, the end date and the status of every membership.
- To record the daily attendance of the members along with the check-in time, the check-out time and the workout duration.

- To allow the stored member information to be searched, updated and deleted by the administrator.
- To generate attendance and membership reports such as the daily attendance, the attendance percentage of a member, the average workout duration and the list of active, expired and expiring memberships.
- To provide all the above facilities through a web browser using HTML, CSS, JavaScript, PHP and MySQL.

CHAPTER II

SYSTEM REQUIREMENT SPECIFICATION

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SYSTEM REQUIREMENT SPECIFICATION

Requirement analysis is the stage of the software development life cycle in which the expectations of the user are collected, examined and written down in a definite form before any code is written. The purpose of this application is the automation of the membership and attendance records of a gymnasium. It provides the administrator of the gym with facilities to register a new member, to assign a membership plan to that member, to mark the attendance of the members on every working day, to modify or remove the stored records and to obtain reports from the information that has been collected.

The system is intended to be used by the administrator of the gymnasium. The application is not open to the general public; the administrator logs in with a user name and a password that are stored in the database, and the operations of the system become available only after the login is successful. The application can be installed in any gymnasium, and the membership plans stored in the database can be changed according to the plans that the particular gymnasium offers.

The technology used for this application is HTML, CSS and JavaScript at the front end, PHP as the server side scripting language and MySQL as the back end. The application is executed on an Apache web server, and the hardware and software required for its development and execution are given below.

2.1 HARDWARE REQUIREMENT

For running the proposed system the following hardware is required:

COMPONENT	SPECIFICATION
Processor	Intel Core i3 or higher
RAM	4 GB (8 GB recommended)
Hard Disk	128 GB or higher
Monitor	Standard colour monitor
Input Devices	Standard keyboard and mouse

2.2 SOFTWARE REQUIREMENT

For running the proposed system the following software is required:

COMPONENT	SPECIFICATION
Operating System	Windows 10 / Windows 11
Front End	HTML, CSS, JavaScript

COMPONENT	SPECIFICATION
Server Side Scripting	PHP
Back End	MySQL
Web Server	Apache (XAMPP package)
Database Language	SQL (Structured Query Language)
IDE / Editor	Visual Studio Code
Web Browser	Google Chrome / Microsoft Edge
Development Assistance	Claude Agent, used only as a coding assistant during development

It may be noted that Claude Agent was used only as an assistant while writing the code of the application. It is not a part of the delivered system, and the application that is submitted is an ordinary PHP and MySQL web application which does not require any such tool in order to run.

CHAPTER III

SYSTEM DESIGN

CHAPTER III

SYSTEM DESIGN

System design is the stage that follows requirement analysis, in which the requirements that have been collected are converted into a plan for the solution. It is in this stage that the movement from the problem to the solution actually takes place, and the decisions taken here affect the quality of the software that is finally produced, because the implementation, the testing and the later maintenance of the system are all carried out on the structure that is fixed during design.

The design of FitTrack is described here in three parts. The database design fixes the tables in which the data of the system is stored, the relationships between those tables and the meaning of every column. The high level design identifies the modules of which the system is composed and states what each module is responsible for. The detailed design then describes, for each module, the inputs that it accepts, the database operation that it performs and the result that it produces.

3.1 DATABASE DESIGN

Database design is the process of producing a detailed model of the data that the system has to store. The model states which tables exist, which columns belong to each table, which column identifies a row uniquely and how a row of one table refers to a row of another table. A design that is prepared carefully at this stage prevents the same piece of information from being stored in more than one place, and therefore prevents the stored information from becoming inconsistent later.

The database of FitTrack is named `fittrack_db` and contains five tables. The members table holds the personal details of the members. The plans table holds the membership plans offered by the gymnasium. The membership table records which member has purchased which plan, and from which date to which date that membership is valid. The attendance table records the visit of a member on a particular date. The admin table holds the login credentials of the administrator of the system.

3.1.1 ER-DIAGRAM

An Entity Relationship diagram is a pictorial representation of the entities of a system, the attributes of those entities and the relationships that exist between them. In the diagram an entity is drawn as a rectangle, an attribute is drawn as an ellipse connected to the entity to which it belongs, and a relationship is drawn as a diamond connecting the entities that take part in it. The attribute that identifies a row of an entity uniquely, that is the primary key, is shown with its name underlined.

The ER diagram of FitTrack consists of five entities, namely admin, members, membership, plans and attendance. A member has a membership, and every membership is based on one of the

plans offered by the gymnasium. The attendance of a member on a particular day is recorded against that member, so that the entity attendance is related to the entity members through the relationship records. The administrator manages the members and, through them, the membership and attendance records of the system.

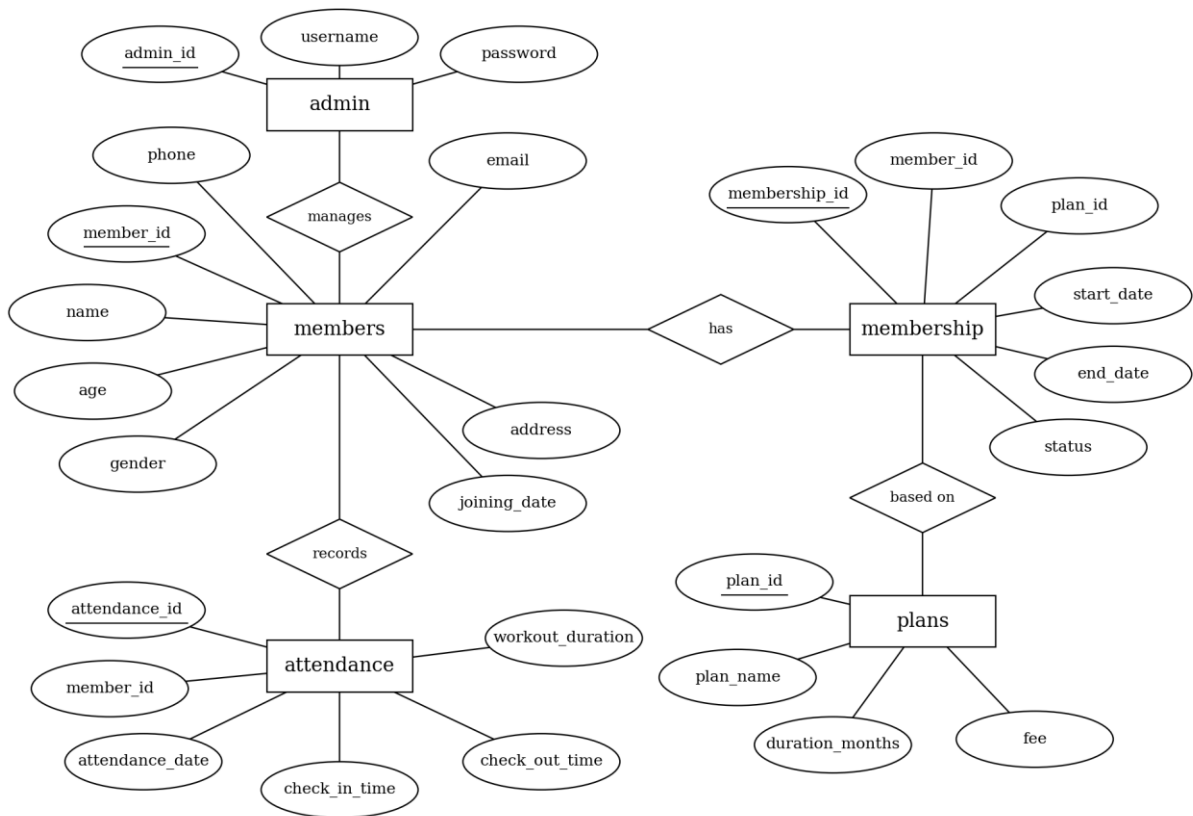


Fig 3.1. ER-DIAGRAM OF FITTRACK: GYM MEMBERSHIP & ATTENDANCE ANALYZER

3.1.2 NORMALIZED TABLES

Normalization is the process of organising the columns of a database into tables in such a way that the same fact is not stored more than once. The process is carried out in a sequence of stages called normal forms. An unnormalized relation is first brought into the first normal form, then into the second normal form and then into the third normal form, and at each stage a particular kind of undesirable dependency is removed.

If all the details of a gymnasium were kept in a single relation, that relation would contain the member details, the plan details and the attendance details together, as shown below. Such a relation is said to be unnormalized, because the details of a member are repeated once for every day on which that member attends the gym, and the details of a plan are repeated once for every member who purchases that plan.

UNNORMALIZED RELATION:

member_id	name	age	gender	phone	plan_name	fee	start_date	end_date	attendance_date	check_in_time	check_out_time
-----------	------	-----	--------	-------	-----------	-----	------------	----------	-----------------	---------------	----------------

FIRST NORMAL FORM (1NF)

A relation is in the first normal form if every column of the relation holds only a single, indivisible value and if the relation contains no repeating group of columns. In the unnormalized relation given above, the attendance of a member is a repeating group, because one member attends the gym on many different dates. This repeating group is removed by placing every visit in a row of its own, so that each row of the relation now contains exactly one attendance date, one check-in time and one check-out time.

SECOND NORMAL FORM (2NF)

A relation is in the second normal form if it is in the first normal form and if every non key column depends on the whole of the primary key and not merely on a part of it. After the first normal form has been reached, the relation is identified by the member identification number together with the attendance date, but the name, the age, the gender and the phone number of the member depend only on the member identification number, and the name and the fee of the plan depend only on the plan. These partial dependencies are removed by separating the member details into the members relation, the plan details into the plans relation and the visits into the attendance relation.

THIRD NORMAL FORM (3NF)

A relation is in the third normal form if it is in the second normal form and if no non key column depends on another non key column. The details of the membership purchased by a member, namely the plan that was purchased and the dates between which it is valid, do not depend on the member alone, because the same member may purchase a further membership when the present one expires. These details are therefore placed in a separate membership relation, which refers to the members relation and to the plans relation through their primary keys. The relations obtained at the end of this process are given below, and the primary key of each relation is shown as the first column.

MEMBERS

member_id	name	age	gender	phone	email	address	joining_date
-----------	------	-----	--------	-------	-------	---------	--------------

PLANS

plan_id	plan_name	duration_months	fee
---------	-----------	-----------------	-----

MEMBERSHIP

membership_id	member_id	plan_id	start_date	end_date	status
---------------	-----------	---------	------------	----------	--------

ATTENDANCE

attendance_id	member_id	attendance_date	check_in_time	check_out_time	workout_duration
---------------	-----------	-----------------	---------------	----------------	------------------

ADMIN

admin_id	username	password
----------	----------	----------

Because the relations are in the third normal form, the anomalies that occur in an unnormalized relation are avoided. Data redundancy is reduced, since the details of a member are written once in the members relation and are referred to from the other relations through the member identification number. The update anomaly is avoided, because a change such as a new phone number is made in exactly one row and is immediately correct everywhere it is used. The insertion anomaly is avoided, because a new membership plan can be entered in the plans relation even though no member has yet purchased it. The deletion anomaly is avoided, because the removal of the attendance of a member does not remove the details of that member, which are held in a relation of their own.

3.1.3 DATA DICTIONARY

A data dictionary is a centralised description of the data that is stored in a database. For every table it gives the name of each column, the type of value that the column holds, the length of that value, the constraint that applies to the column and the meaning of the column. The data dictionary of the five tables of FitTrack is given below.

Table: MEMBERS

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
1	member_id	INT	11	Primary Key	Identification number of the member, generated automatically
2	name	VARCHAR	50	Not Null	Name of the member
3	age	INT	3	Not Null	Age of the member in completed years
4	gender	VARCHAR	10	Not Null	Gender of the member
5	phone	VARCHAR	15	Not Null	Contact number of the member
6	email	VARCHAR	50	-	Email address of the member
7	address	VARCHAR	150	-	Residential address of the member
8	joining_date	DATE	-	Not Null	Date on which the member joined the gymnasium

Table: PLANS

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
1	plan_id	INT	11	Primary Key	Identification number of the membership plan
2	plan_name	VARCHAR	30	Not Null	Name of the membership plan
3	duration_months	INT	3	Not Null	Validity of the plan expressed in months
4	fee	DECIMAL	10,2	Not Null	Fee payable for the plan

Table: MEMBERSHIP

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
1	membership_id	INT	11	Primary Key	Identification number of the membership record
2	member_id	INT	11	Foreign Key	Refers to member_id of the members table
3	plan_id	INT	11	Foreign Key	Refers to plan_id of the plans table
4	start_date	DATE	-	Not Null	Date on which the membership begins

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
5	end_date	DATE	-	Not Null	Date on which the membership expires
6	status	VARCHAR	15	Not Null	Status of the membership: Active, Expired or Expiring Soon

Table: ATTENDANCE

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
1	attendance_id	INT	11	Primary Key	Identification number of the attendance record
2	member_id	INT	11	Foreign Key	Refers to member_id of the members table
3	attendance_date	DATE	-	Not Null	Date on which the member attended the gymnasium
4	check_in_time	TIME	-	Not Null	Time at which the member entered the gymnasium
5	check_out_time	TIME	-	-	Time at which the member left the gymnasium
6	workout_duration	INT	5	-	Duration of the workout in minutes, obtained from the check-in and check-out time

Table: ADMIN

S.NO	COLUMN NAME	DATA TYPE	LENGTH	CONSTRAINT	DESCRIPTION
1	admin_id	INT	11	Primary Key	Identification number of the administrator
2	username	VARCHAR	30	Not Null	User name used to log into the application
3	password	VARCHAR	255	Not Null	Password used to log into the application

3.2 HIGH LEVEL DESIGN

A high level design gives an overview of the system by stating the modules of which it is composed and the responsibility of each module. It does not describe how a module works internally; it describes only what the module is expected to do, so that the modules can be developed separately and can later be put together. FitTrack consists of the following five modules.

MEMBER REGISTRATION MODULE

In the member registration module the administrator enters the personal details of a person who wishes to join the gymnasium. The details entered are the name, the age, the gender, the phone number, the email address, the residential address and the joining date. The values entered in the form are checked before they are stored, and the accepted details are inserted as a new row of the members table. A member identification number is generated for the new row, and this number is used to refer to that member in the remaining modules of the system.

MEMBERSHIP MANAGEMENT MODULE

In the membership management module a membership plan is assigned to a member who has already been registered. The administrator selects the member and the plan, and the start date and the end date of the membership are recorded in the membership table. The module also displays the membership details of a member and allows an existing membership to be updated when the member renews it. The status of the membership is maintained as Active while the current date lies within the validity of the membership, as Expiring Soon when the end date is near and as Expired once the end date has passed.

ATTENDANCE MANAGEMENT MODULE

In the attendance management module the visit of a member to the gymnasium is recorded. The administrator selects the member and enters the attendance date, the check-in time and the check-out time, and these values are inserted as a new row of the attendance table. The duration of the workout is obtained from the check-in time and the check-out time and is stored along with the attendance record, so that the duration need not be calculated again whenever a report is prepared.

MEMBER MANAGEMENT MODULE

In the member management module the records that are already present in the database are maintained. The administrator searches for a member, and the details of that member are displayed in a form. The details that have changed are modified and the modified values replace the earlier values in the members table. When a member record is no longer required, it is removed from the database, and the membership and attendance records that belong to that member are removed along with it so that no record is left referring to a member who does not exist.

REPORT AND ANALYSIS MODULE

In the report and analysis module the information that has been collected by the other modules is read back and presented to the administrator. The module displays the attendance of all the members on a particular date, the attendance of an individual member over a period, the attendance percentage of a member, the average workout duration of a member and the lists of memberships that are active, that have expired and that are about to expire. The attendance percentage and the average workout duration are obtained by the formulae given in the detailed design.

In addition to the five modules described above, the application provides an administrator login through which the system is protected. The user name and the password entered on the login page are compared with the values stored in the admin table, and the modules of the system become available only when the values match.

3.3 DETAILED DESIGN

The detailed design describes each module of the system individually. Since the application is written in PHP, a module is implemented as a PHP page rather than as a function of a programming language. For every module the following description therefore gives the values that the page receives, the value that the page produces, the work that the page carries out, the page from which it is reached and the pages that it reaches in turn. The database operation performed by each module is also stated.

Module: Administrator Login

Module name	Administrator Login
Parameters	User name and password submitted from the login form
Return value	A session for the administrator when the credentials match, otherwise an error message
Description	Reads the admin table with a SELECT query and compares the entered user name and password with the stored values. When they match, a session variable is set and the administrator is taken to the main menu; when they do not match, the login page is displayed again with an error message.
Database operation	SELECT
Calling function	-
Function called by module	Main menu

Module: Main Menu

Module name	Main Menu
Parameters	None
Return value	None
Description	Displays the operations available in the system and waits for the administrator to choose one of them, after which the page belonging to the chosen operation is opened.
Database operation	-
Calling function	Administrator Login
Function called by module	Member Registration, Membership Management, Attendance Management, Member Management, Report and Analysis

Module: Member Registration

Module name	Member Registration
Parameters	Name, age, gender, phone number, email, address and joining date submitted from the registration form
Return value	A confirmation message containing the member identification number that has been generated

Description	Checks the values submitted in the form and inserts them as a new row of the members table. The member identification number is generated automatically for the new row and is displayed to the administrator so that it can be used in the other modules.
Database operation	INSERT
Calling function	Main menu
Function called by module	Membership Management

Module: Membership Management

Module name	Membership Management
Parameters	Member identification number, plan identification number, start date and end date
Return value	The membership details of the selected member
Description	Assigns a plan to a registered member by inserting a row into the membership table, displays the membership details of a member by reading the membership table together with the members table and the plans table, and updates an existing membership when it is renewed. The status of the membership is obtained by comparing the end date with the current date.
Database operation	INSERT, SELECT and UPDATE
Calling function	Main menu, Member Registration
Function called by module	Report and Analysis

Module: Attendance Management

Module name	Attendance Management
Parameters	Member identification number, attendance date, check-in time and check-out time
Return value	A confirmation message and the attendance of the selected date
Description	Inserts the visit of a member as a new row of the attendance table. The workout duration is obtained as the difference between the check-out time and the check-in time and is stored in the same row. The attendance that has been marked on the selected date is then displayed.
Database operation	INSERT
Calling function	Main menu
Function called by module	Report and Analysis

Module: Member Management

Module name	Member Management
Parameters	Member identification number, and the member details that have been modified
Return value	A confirmation message stating that the record has been updated or removed

Description	Searches for a member and displays the stored details in a form. When the modified details are submitted, the corresponding row of the members table is updated. When the record is to be removed, the row of the members table is deleted along with the membership and attendance rows that refer to that member.
Database operation	SELECT, UPDATE and DELETE
Calling function	Main menu
Function called by module	-

Module: Report and Analysis

Module name	Report and Analysis
Parameters	The date or the member identification number for which the report is required
Return value	The report displayed as a table on the page
Description	Reads the attendance table and the membership table and presents the daily attendance, the attendance of an individual member, the attendance percentage, the average workout duration and the lists of active, expired and expiring memberships. The attendance percentage is obtained as the number of days present divided by the total number of working days, multiplied by one hundred, and the average workout duration is obtained as the total workout duration divided by the total number of attendance days.
Database operation	SELECT
Calling function	Main menu
Function called by module	-

CHAPTER IV

IMPLEMENTATION

CHAPTER IV

IMPLEMENTATION

Implementation is the stage in which the design that has been prepared is converted into a working application. The pages of FitTrack are written in HTML and are formatted using CSS. The values entered in a form are checked in the browser using JavaScript before the form is submitted. The submitted values are received by a PHP page on the server, and that page connects to the MySQL server and executes an SQL query in order to insert, read, modify or remove the data. The result produced by the query is then written back into an HTML page and is returned to the browser, which displays it to the administrator.

The communication with the database is carried out through the MySQLi extension of PHP. A single connection file is written and is included at the beginning of every page that needs the database, so that the details of the connection occur in only one place and can be changed easily when the application is moved to another computer. The values that are received from a form are passed to the query through prepared statements, so that the value entered by the user is treated only as a value and not as a part of the SQL statement.

The administrator login is implemented using PHP sessions. When the user name and the password entered on the login page are found in the admin table, a session variable is created, and every other page of the application checks for that session variable before it displays anything. When the session variable is not present, the browser is redirected to the login page, so that the modules of the system cannot be reached without logging in. The logout page destroys the session.

The reports are produced by SELECT queries. The attendance of a date is obtained by reading the attendance table for that date together with the members table so that the name of the member can be displayed. The attendance percentage and the average workout duration of a member are obtained by counting the attendance rows of that member and by taking the average of the workout duration column. The membership status is obtained by comparing the end date stored in the membership table with the current date.

4.1 DEPLOYMENT PROCEDURE

The application is deployed on a computer on which the XAMPP package has been installed. The steps to be followed are given below.

- Install the XAMPP package, which provides the Apache web server, the PHP interpreter and the MySQL database server.
- Start the Apache module and the MySQL module from the XAMPP control panel.
- Copy the fittrack project folder into the htdocs directory of the XAMPP installation.

- Open phpMyAdmin in the browser, create a database with the name fittrack_db and import the fittrack_db.sql file supplied with the project, so that the five tables are created and the initial rows are inserted.
- Open the database connection file of the project and verify that the host name, the user name, the password and the database name agree with the MySQL server that has been started.
- Open the browser and enter the address <http://localhost/fittrack> in order to reach the login page of the application.
- Log in with the administrator user name and password stored in the admin table.
- Test the modules one after another by registering a member, assigning a membership, marking the attendance, updating and deleting a record and generating the reports.

4.2 CODING

The listings given in this section show the parts of the code that are representative of each module of the application. The complete source code of the project is submitted separately along with this report.

Database connection

```
<?php
// db_connect.php - connection used by every page of the application
$host = "localhost";
$user = "root";
$pass = "";
$dbname = "fittrack_db";

$conn = new mysqli($host, $user, $pass, $dbname);
if ($conn->connect_error) {
    die("Connection failed : " . $conn->connect_error);
}
?>
```

Member registration (INSERT)

```
<?php
// register_member.php - Module 1
include("db_connect.php");

if (isset($_POST["submit"])) {
    $name = trim($_POST["name"]);
    $age = (int) $_POST["age"];
    $gender = $_POST["gender"];
    $phone = trim($_POST["phone"]);
    $email = trim($_POST["email"]);
    $address = trim($_POST["address"]);
    $jdate = $_POST["joining_date"];

    $sql = "INSERT INTO members
            (name, age, gender, phone, email, address, joining_date)
            VALUES (?, ?, ?, ?, ?, ?, ?)";

    $stmt = $conn->prepare($sql);
    $stmt->bind_param("sisssss", $name, $age, $gender, $phone,
                      $email, $address, $jdate);

    if ($stmt->execute()) {
        $id = $conn->insert_id;
        echo "Member registered successfully. Member ID : " . $id;
    } else {
        echo "Registration failed. Please try again.";
    }
    $stmt->close();
}
?>
```

Membership management (INSERT and status)

```

<?php
// add_membership.php - Module 2
include("db_connect.php");

if (isset($_POST["assign"])) {
    $member_id = (int) $_POST["member_id"];
    $plan_id    = (int) $_POST["plan_id"];
    $start      = $_POST["start_date"];

    // end date is obtained from the duration of the selected plan
    $p = $conn->prepare("SELECT duration_months FROM plans
                        WHERE plan_id = ?");
    $p->bind_param("i", $plan_id);
    $p->execute();
    $p->bind_result($months);
    $p->fetch();
    $p->close();

    $end = date("Y-m-d", strtotime($start . " + $months month"));
    $status = (strtotime($end) >= strtotime(date("Y-m-d")))
        ? "Active" : "Expired";

    $sql = "INSERT INTO membership
            (member_id, plan_id, start_date, end_date, status)
            VALUES (?, ?, ?, ?, ?)";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param("iisss", $member_id, $plan_id, $start,
        $end, $status);
    $stmt->execute();
    $stmt->close();
    echo "Membership assigned. Valid up to " . $end;
}
?>

```

Attendance marking (INSERT with workout duration)

```

<?php
// mark_attendance.php - Module 3
include("db_connect.php");

if (isset($_POST["mark"])) {
    $member_id = (int) $_POST["member_id"];
    $adate      = $_POST["attendance_date"];
    $in         = $_POST["check_in_time"];
    $out        = $_POST["check_out_time"];

    // workout duration in minutes
    $duration = (strtotime($out) - strtotime($in)) / 60;

    $sql = "INSERT INTO attendance
            (member_id, attendance_date, check_in_time,
             check_out_time, workout_duration)
            VALUES (?, ?, ?, ?, ?)";

    $stmt = $conn->prepare($sql);
    $stmt->bind_param("isssi", $member_id, $adate, $in,
        $out, $duration);
    $stmt->execute();
    $stmt->close();
    echo "Attendance marked. Workout duration : " .
        $duration . " minutes";
}

```

```
}
?>
```

Updating and deleting a member (UPDATE and DELETE)

```
<?php
// update_member.php - Module 4
include("db_connect.php");
if (isset($_POST["update"])) {
    $id = (int) $_POST["member_id"];
    $name = trim($_POST["name"]);
    $phone = trim($_POST["phone"]);
    $addr = trim($_POST["address"]);
    $sql = "UPDATE members SET name = ?, phone = ?, address = ?
        WHERE member_id = ?";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param("sssi", $name, $phone, $addr, $id);
    $stmt->execute();
    $stmt->close();
    echo "Member details updated successfully.";
}
if (isset($_GET["delete"])) {
    $id = (int) $_GET["delete"];
    $conn->query("DELETE FROM attendance WHERE member_id = $id");
    $conn->query("DELETE FROM membership WHERE member_id = $id");
    $conn->query("DELETE FROM members WHERE member_id = $id");
    echo "Member record removed from the database.";
}
?>
```

Report and analysis (SELECT with calculation)

```
<?php
// report.php - Module 5
include("db_connect.php");

$member_id = (int) $_GET["member_id"];
$working_days = (int) $_GET["working_days"];

$sql = "SELECT COUNT(*) AS present_days,
        AVG(workout_duration) AS avg_duration
        FROM attendance
        WHERE member_id = ?";
$stmt = $conn->prepare($sql);
$stmt->bind_param("i", $member_id);
$stmt->execute();
$res = $stmt->get_result();
$row = $res->fetch_assoc();

$present = (int) $row["present_days"];
$avg = round($row["avg_duration"], 2);

// Attendance Rate = (Present Days / Total Working Days) * 100
$percentage = ($working_days > 0)
    ? round(($present / $working_days) * 100, 2) : 0;

echo "Days present          : " . $present;
echo "Attendance rate      : " . $percentage . " %";
echo "Average workout      : " . $avg . " minutes";
$stmt->close();
?>
```

4.3 SCREENSHOTS

The screens of the application obtained while the modules were executed are given below.

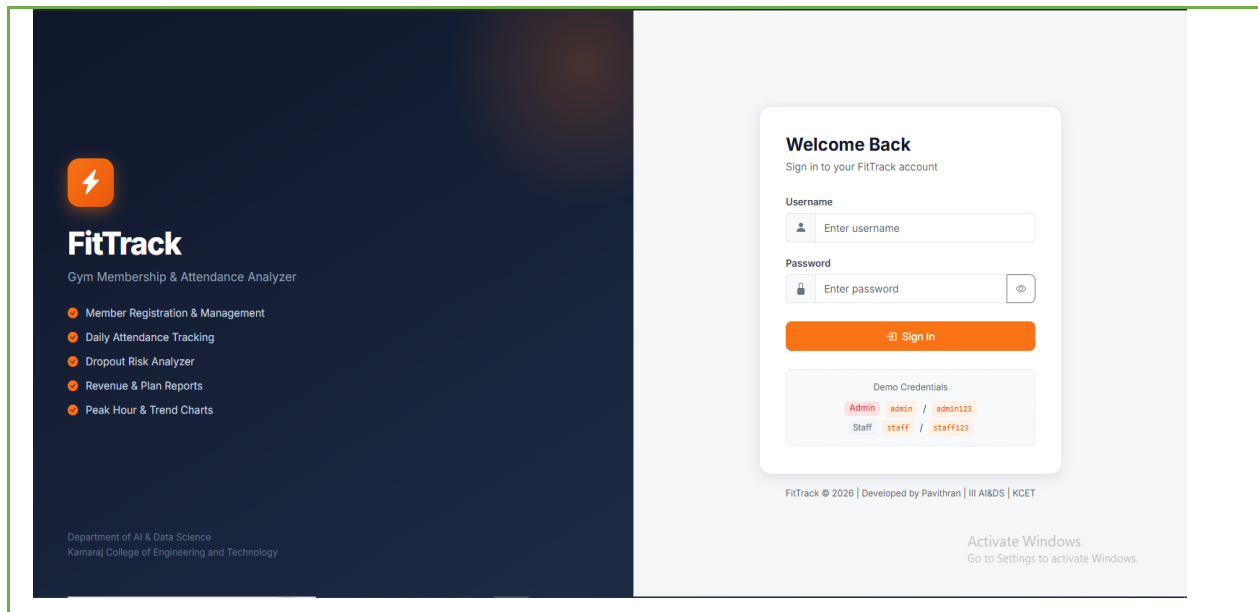


Fig 4.1. FITTRACK ADMINISTRATOR LOGIN PAGE

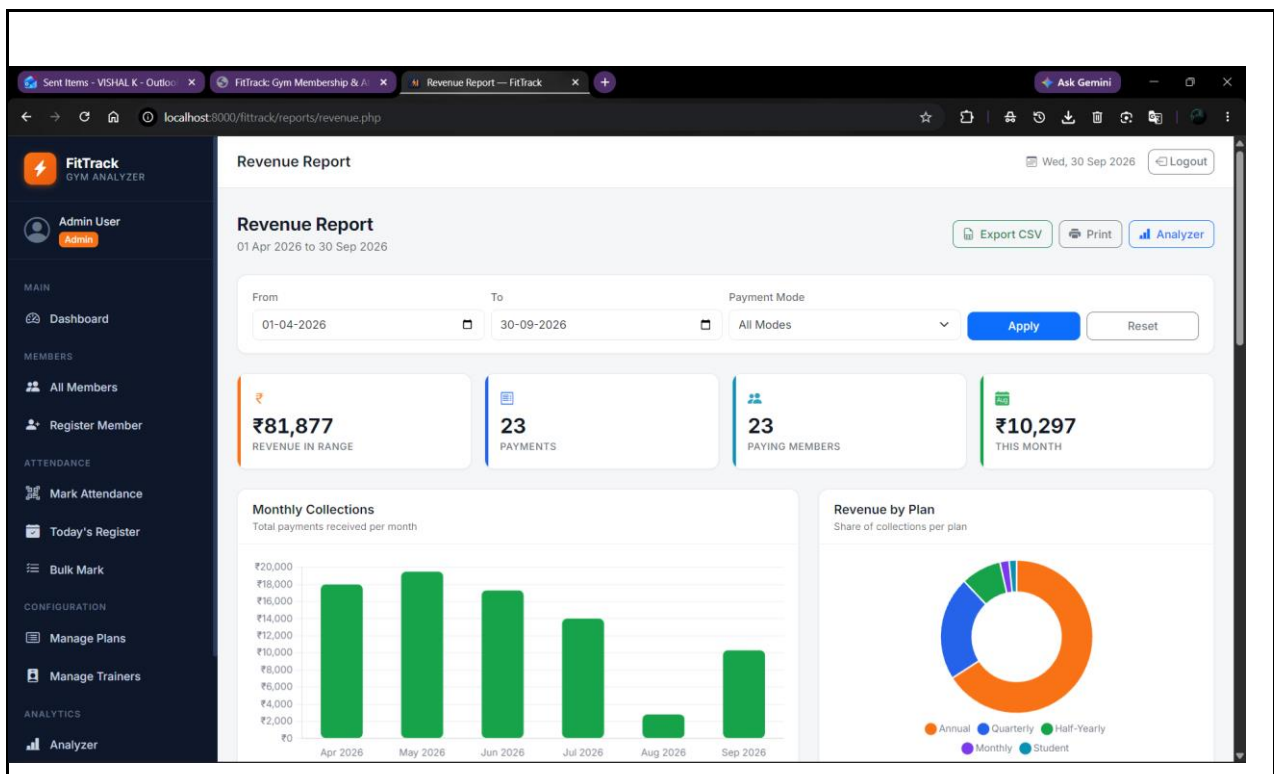


Fig 4.2. FITTRACK MAIN MENU

FitTrack GYM ANALYZER

Admin User Admin

MAIN

- Dashboard

MEMBERS

- All Members
- Register Member**

ATTENDANCE

- Mark Attendance
- Today's Register
- Bulk Mark

CONFIGURATION

- Manage Plans
- Manage Trainers

ANALYTICS

- Analyzer

Register Member

Wed, 30 Sep 2026 Logout

Register New Member
Module 1 — INSERT Operation

← Back to List

Personal Details

Full Name * Vishal Gender * Male Date of Birth * dd-mm-yyyy

Phone (10 digits) * 6374712490 Email Address k.vishal108737@gmail.com

Address 22, Gandhi Rd, Mani Nagar Sivakasi Emergency Contact 9842827621

Membership

Join Date * 30-09-2026

Membership Plan * Student — ₹1,299 (3 months)

Expiry: 30 Dec 2026

Assign Trainer Arjun Selvam (Strength)

Payment Mode * Cash

Register Member

FitTrack © 2026 | Developed by Vishal | III AI&DS | KCET

Fig 4.3. MEMBER REGISTRATION

FitTrack GYM ANALYZER

Admin User Admin

MAIN

- Dashboard

MEMBERS

- All Members**
- Register Member

ATTENDANCE

- Mark Attendance
- Today's Register
- Bulk Mark

CONFIGURATION

- Manage Plans
- Manage Trainers

ANALYTICS

- Analyzer

All Members

Wed, 30 Sep 2026 Logout

32 members found

Register New

Search Name, code, or phone... Status All Status Plan All Plans Trainer All Trainers Filter Reset

#	CODE	NAME	PHONE	PLAN	TRAINER	EXPIRY	ATTENDANCE	STATUS	ACTIONS
1	FT-2026-032	BALAYOGHI N Male	9345548984	Annual	Priya Annamalai	30 Sep 2027	100%	Active	
2	FT-2026-031	Vishal Male	6374712490	Monthly	Arjun Selvam	30 Oct 2026	100%	Active	
3	FT-2026-030	Eswari Lakshmi Female	9500001030	Monthly	Deepika Rajan	15 Sep 2026	50%	Expired	
4	FT-2026-029	Durai Murugan Male	9500001029	Quarterly	Arjun Selvam	03 Oct 2026	31.15%	Active	
5	FT-2026-028	Chitra Kannan Female	9500001028	Annual	Priya Annamalai	20 Jul 2027	76.71%	Active	
6	FT-2026-027	Brindha Devi Female	9500001027	Quarterly	Karthick Murugan	05 Oct 2026	44.58%	Active	
7	FT-2026-026	Anbu Selvan Male	9500001026	Annual	Deepika Rajan	01 Jul 2027	81.52%	Active	

Fig 4.4. MEMBER LIST

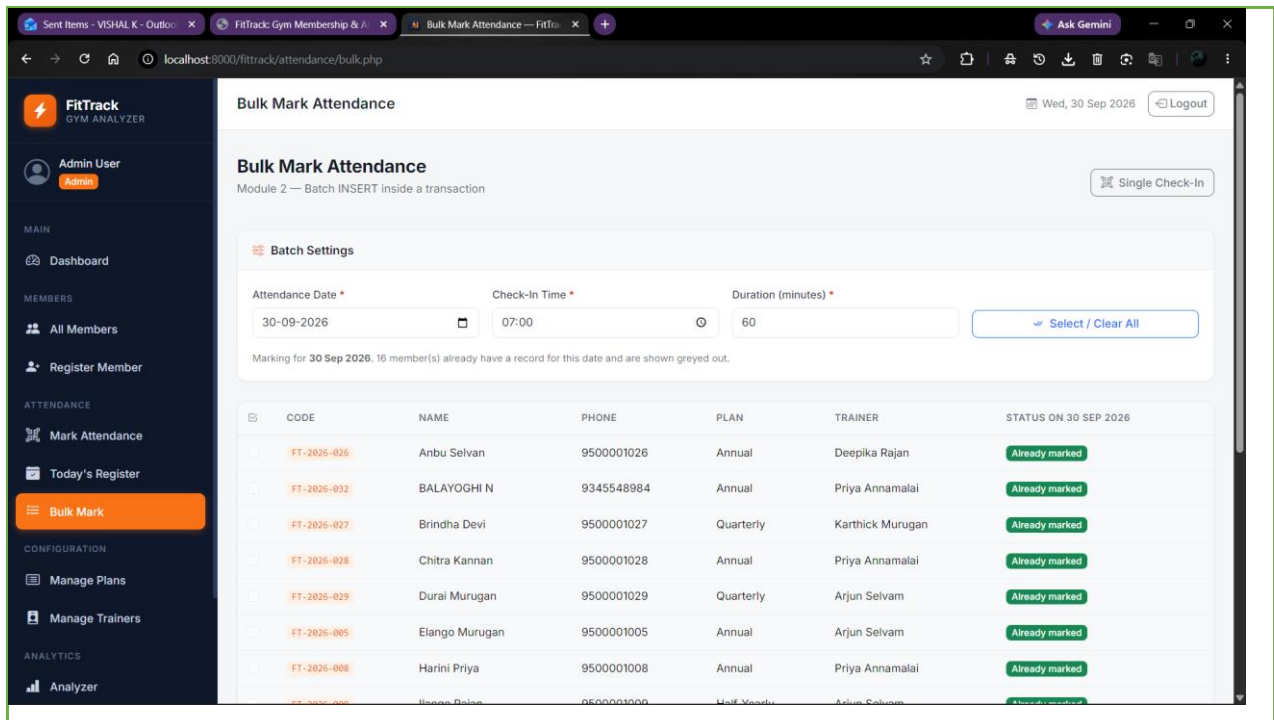


Fig 4.6. ATTENDANCE HISTORY

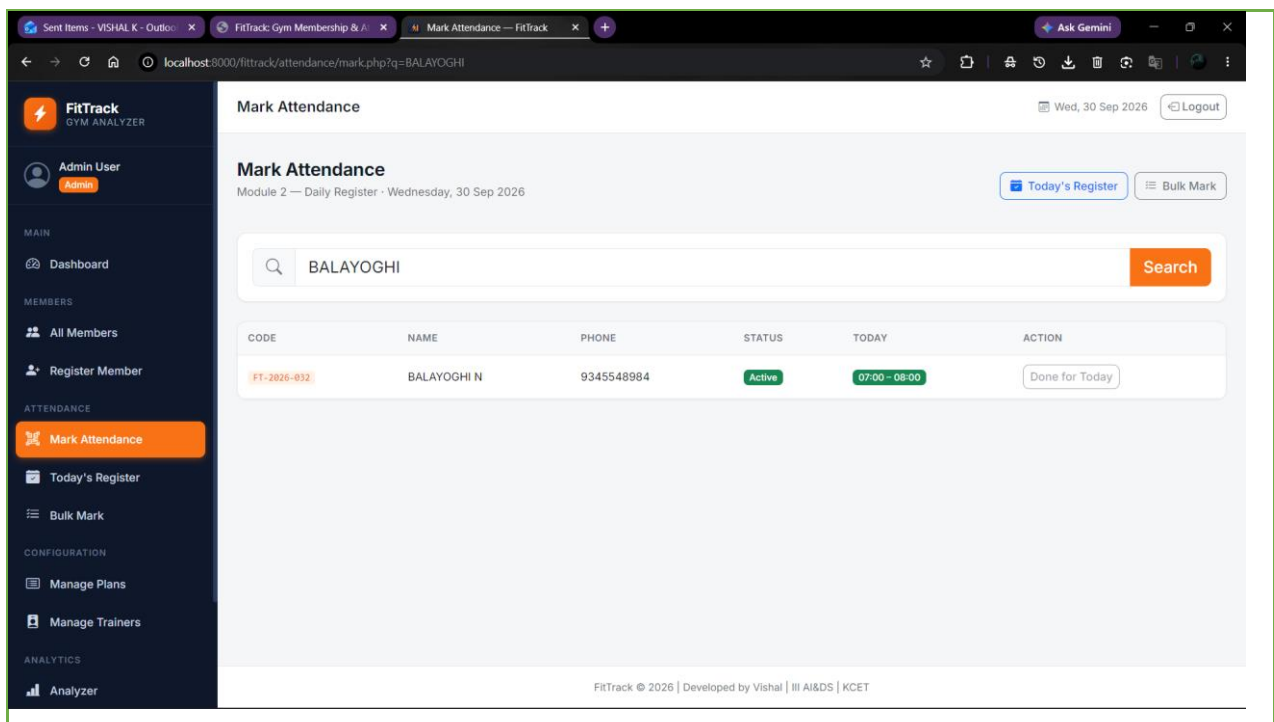


Fig 4.7. ATTENDANCE MARKING

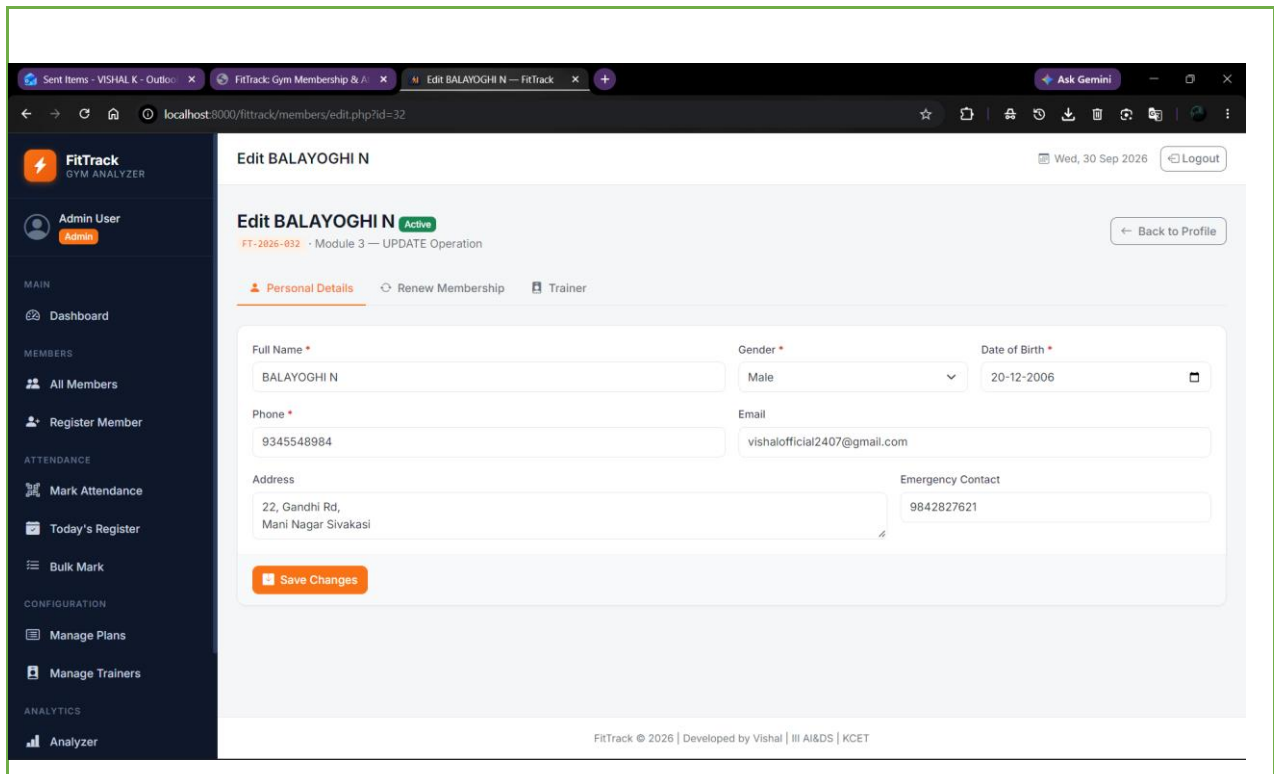


Fig 4.8. UPDATING AND DELETING A MEMBER

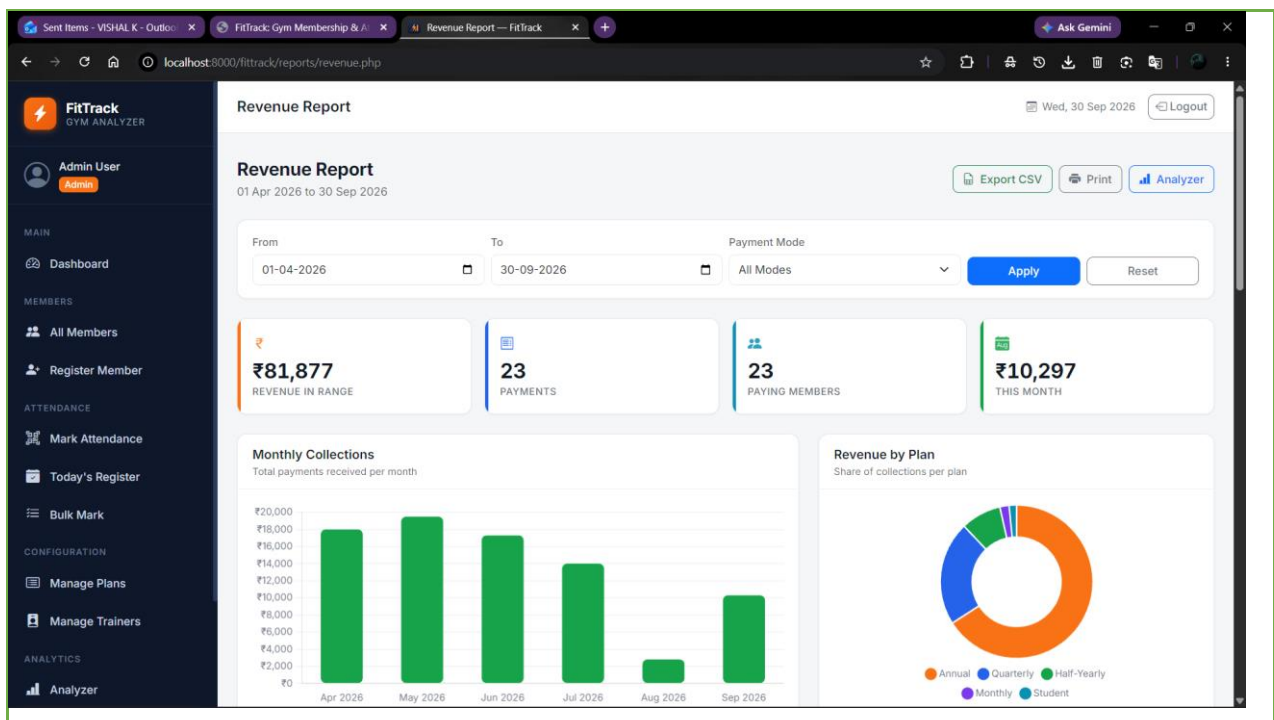


Fig 4.9. ATTENDANCE AND MEMBERSHIP REPORT

```

C:\Users\kvish>cd /d "D:\IP MINI PROJECT\fittrack\sql"
D:\IP MINI PROJECT\fittrack\sql>mysql -h 127.0.0.1 -P 3307 --protocol=TCP -u root --table < demo_queries.sql
+-----+
| Total Registered Members |
+-----+
| 32 |
+-----+
+-----+
| Status | Members |
+-----+
| Active | 16 |
| Expired | 13 |
| Cancelled | 3 |
+-----+
+-----+
| Plan | Members |
+-----+
| Quarterly | 12 |
| Annual | 11 |
| Monthly | 4 |
| Half-Yearly | 3 |
| Student | 2 |
| VISHAL | 0 |
+-----+
+-----+
| Month Registered | New Members |
+-----+
| Feb 2026 | 5 |
| Mar 2026 | 5 |
| Apr 2026 | 5 |
| May 2026 | 5 |
| Jun 2026 | 5 |
| Jul 2026 | 3 |
| Aug 2026 | 2 |
| Sep 2026 | 2 |
+-----+
+-----+
| Code | Name | Joined | Expires | Status |
+-----+
| FT-2026-032 | BALAYOGHI N | 2026-09-30 | 2027-09-30 | Active |
| FT-2026-031 | Vishal | 2026-09-30 | 2026-10-30 | Active |
| FT-2026-030 | Eswari Lakshmi | 2026-08-15 | 2026-09-15 | Expired |
| FT-2026-029 | Durai Murugan | 2026-08-01 | 2026-10-03 | Active |
+-----+

```

Fig 4.10. DATABASE TABLES OF FITTRACK IN phpMyAdmin

CHAPTER V

CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER V

CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The project entitled FitTrack: Gym Membership & Attendance Analyzer has been developed and executed successfully using HTML, CSS and JavaScript at the front end, PHP as the server side scripting language and MySQL as the back end. The five modules of the system, namely member registration, membership management, attendance management, member management and report and analysis, have been implemented, and the four database operations of insertion, selection, updation and deletion are performed through them.

The system stores the details of the members, the membership plans, the membership purchased by each member and the daily attendance in a single database whose tables are in the third normal form. Because of this, the details of a member are stored only once and are referred to from the other tables through the member identification number, and the inconsistency that arises when the same information is written in several registers is avoided. The reports produced by the system give the administrator the daily attendance, the attendance percentage of a member, the average workout duration and the lists of active, expired and expiring memberships, which could not be obtained from a manual register without counting the entries by hand.

The present version of the system is intended to be used by the administrator of the gymnasium from a single computer on which the server is running. The attendance is entered by the administrator, and the members do not have an account of their own through which they can see their attendance. The fees of the memberships are recorded as a part of the plan, and payments are not collected through the application.

5.2 FUTURE ENHANCEMENT

The following enhancements are proposed for the future versions of the system. They are not implemented in the present version.

- QR based attendance, in which a member is identified by scanning a code at the entrance so that the attendance is marked without the administrator having to enter it.
- Online membership renewal, through which a member can renew an expiring membership without visiting the office of the gymnasium.
- Online payment integration, so that the fee of a membership can be collected through the application and a receipt can be generated automatically.

- A mobile application, through which a member can view the membership validity and the attendance record from a mobile phone.
- Advanced attendance analytics, such as the identification of the members who have stopped attending the gymnasium and the study of the hours of the day at which the gymnasium is most crowded.
- A trainer module, in which a trainer is assigned to a member and the workouts prescribed by the trainer are recorded.